



Lab 10 - Network Address Translation (NAT)

What is NAT

Network Address Translation (NAT) is a mechanism that allows a router to modify the IP address information in packet headers while they are in transit. Its primary purpose is to enable multiple devices on a private network to share one or more public IP addresses when communicating with external networks such as the Internet. NAT was introduced as a practical solution to the exhaustion of IPv4 addresses. Without it, every device on a network would require its own globally routable IP address, a requirement that quickly becomes infeasible at scale. By translating private addresses to a smaller pool of public addresses, NAT allows thousands of devices to share a handful of public IPs.

Private vs Public IP Addresses

The Internet Assigned Numbers Authority (IANA) has reserved three ranges of IP addresses exclusively for private (internal) use. These addresses are not routable on the public Internet and any packet carrying a private source address will be dropped by Internet routers. The range is described in the following table:

Address Range	Default Subnet Mask	Address Class
10.0.0.0 – 10.255.255.255	255.0.0.0 (/8)	Class A
172.16.0.0 – 172.31.255.255	255.240.0.0 (/12)	Class B
192.168.0.0 – 192.168.255.255	255.255.0.0 (/16)	Class C

Public IP addresses, by contrast, are globally unique and routable on the Internet. They are assigned by Internet Service Providers (ISPs) and managed by regional registries. Organizations typically own far fewer public addresses than they have internal devices, which is precisely where NAT bridges the gap.

How NAT Works

When a device on the internal network sends a packet destined for the Internet, the NAT-enabled router intercepts it and replaces the source private IP address with a public IP address before forwarding the packet. The router also records this mapping in a NAT translation table. When a reply arrives from the external host, the router consults the table, reverses the translation, and forwards the packet to the correct internal device. This process is transparent to both endpoints. The internal device believes it is communicating directly with the remote server, and the remote server only ever sees the router's public address.



Types of NAT

There are different types of NAT that are used in networks, the details of which are described as follows.

Static NAT

Static NAT creates a fixed, one-to-one mapping between a single private IP address and a single public IP address. This mapping persists permanently in the NAT table and is manually configured by the administrator. Static NAT is commonly used to expose internal servers (web, mail, FTP) to the Internet under a specific public address.

Use case: A company hosts an internal web server at 192.168.1.10 and wants it accessible from the Internet as 203.0.113.10. Static NAT creates this permanent binding.

In static NAT:

- One private address maps to exactly one public address
- The mapping is permanent and manually configured
- Requires one public IP per internally mapped device

Dynamic NAT

Dynamic NAT maps private IP addresses to public addresses drawn from a pre-configured pool of available public IPs. Unlike static NAT, the mapping is not fixed. The router assigns a free public address from the pool at the time a session is initiated and releases it once the session ends.

Use case: An organization has 20 public IP addresses and 80 internal hosts. At any given time, no more than 20 hosts need Internet access simultaneously. Dynamic NAT can serve all 80 hosts using just those 20 public IPs on a first come, first served basis.

In dynamic NAT:

- There is a pool of public IP addresses
- Mappings are created dynamically and expire after a timeout period
- External hosts cannot initiate connections (no predictable public address)
- If the pool is exhausted, new sessions are dropped

Port Address Translation (PAT) also known as NAT Overload

PAT extends the concept of dynamic NAT by also translating the transport-layer port numbers, which allows many private addresses to share a single public IP address simultaneously. Each connection is distinguished not only by IP address but by a unique port number. PAT is the form of NAT used in virtually all home routers and most enterprise edge routers.



Use case: A home router connects dozens of devices to the Internet using a single public IP address assigned by the ISP. PAT tracks every connection by its unique combination of source IP, source port, and protocol.

In PAT:

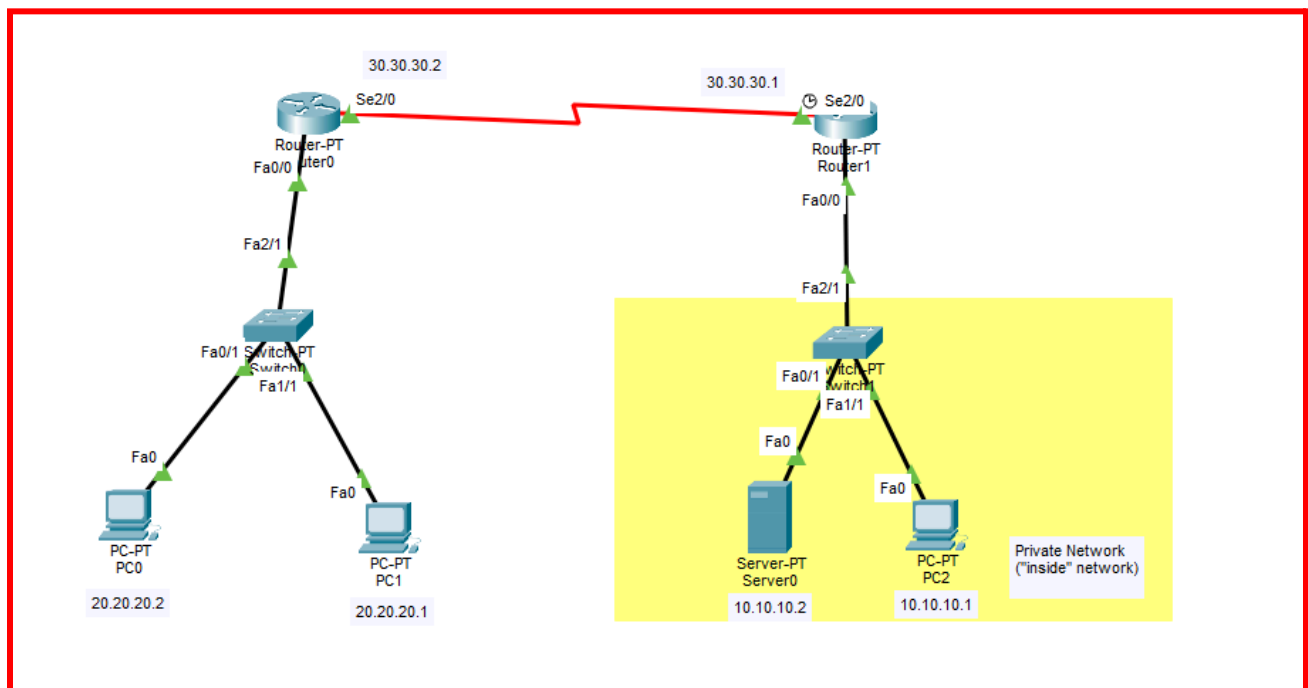
- Many-to-one mapping (all hosts share a single public IP)
- Port numbers are used to uniquely identify each session
- Supports tens of thousands of simultaneous sessions on one public IP

Lab Tasks

Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).

CLO2: Apply core networking concepts and protocols, including but not limited to routing, Congestion control, and error management, to establish and analyze reliable data communication across networks.

Task 1: Static NAT (10 marks)





Step 1: Setting Up the Network Topology

Open Cisco Packet Tracer and place two routers (Router-PT) on the workspace, the one facing the public network will be the ISP Router and the one facing the internal network will be the Internal Router. Use a Serial DCE cable to connect the two routers together, as this represents a wide-area network (WAN) link where the clock rate is typically controlled by the service provider. Next, add a switch (Switch-PT) to the internal side and connect a PC and a Server (Server-PT) to it using copper straight-through cables. On the public side, connect another switch (Switch-PT) to the router and attach two additional PCs to represent users on the internet. Make sure to follow the ports as shown in the diagram above.

Step 2: Configuring IP Addresses and Connectivity

Before NAT can function, each device must have a valid IP configuration. For the private network, assign the PC an IP of 10.10.10.1 and the Server an IP of 10.10.10.2, both with a subnet mask of 255.0.0.0 and a default gateway of 10.10.10.0. On the Server, also turn on HTTP/HTTPS services and then edit the index.html file. Add a <h1> element that says your **name, roll number, section and "lab number 10"**, to it. On the Internal Router, configure the interface facing the switch with the IP 10.10.10.0 and turn the port status on (go to config > then the corresponding interface). On the external Router, configure the interface facing the switch with the IP 20.20.20.0 and turn the port status on (go to config > then the corresponding interface). For the connection between routers, assign the Internal Router's serial interface an IP of 30.30.30.1 (go to config > then the corresponding interface) and the ISP Router's serial interface 30.30.30.2 (go to config > then the corresponding interface). Finally, configure the public network PCs with IP addresses 20.20.20.1 and 20.20.20.2, the default gateway for both is 20.20.20.0.

Step 3: Implementing Static NAT rules

The core of this task involves creating the static translation on the Internal Router. Access the Command Line Interface (CLI) and enter global configuration mode. You will use the command `ip nat inside source static` followed by the private IP and then the public IP you wish to map it to. For example, mapping the PC would require the command `ip nat inside source static 10.10.10.1 30.30.30.1`. After defining these mappings, you must tell the router which interfaces are part of the "inside" network and which are "outside." Navigate to the interface connected to your switch and apply the command `ip nat inside`, then navigate to the serial interface connected to the ISP and apply `ip nat outside`.

To achieve this, follow these commands.

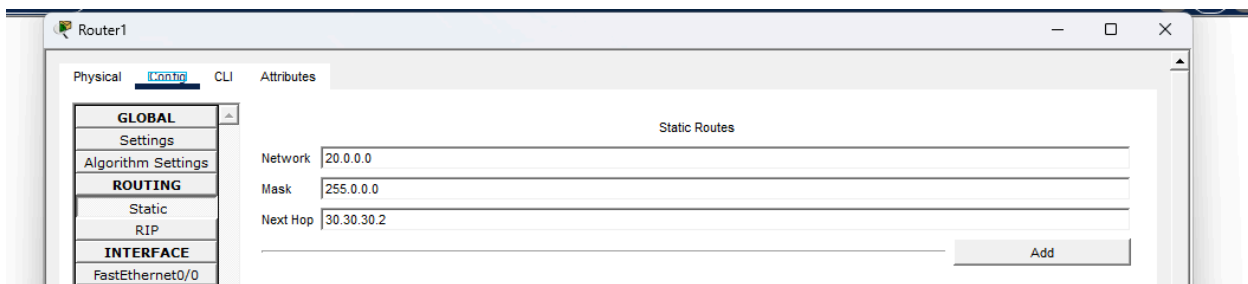
Open **Router1 (the internal router)** and go to the **CLI** tab. Run the following commands.



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```
Router(config-if)#exit
Router(config)#ip nat inside source static 10.10.10.1 30.30.30.1 //maps PC2 to
the public address
Router(config)#ip nat inside source static 10.10.10.2 30.30.30.1 //maps the
Server to the public address
Router(config)#int fa0/0
Router(config-if)#ip nat inside //the side connected to the switch is inside the
network
Router(config-if)#int s2/0
Router(config-if)#ip nat outside //the side connected to the second router is
outside the network
Router(config-if)#exit
```

Then go to the Config tab > Static, and enter the following details then click on “Add”.



Here you are adding a static route to the internal router so it can find the public network. This will route all public traffic to the router0 (the external router), enabling communication between the public and the private network.

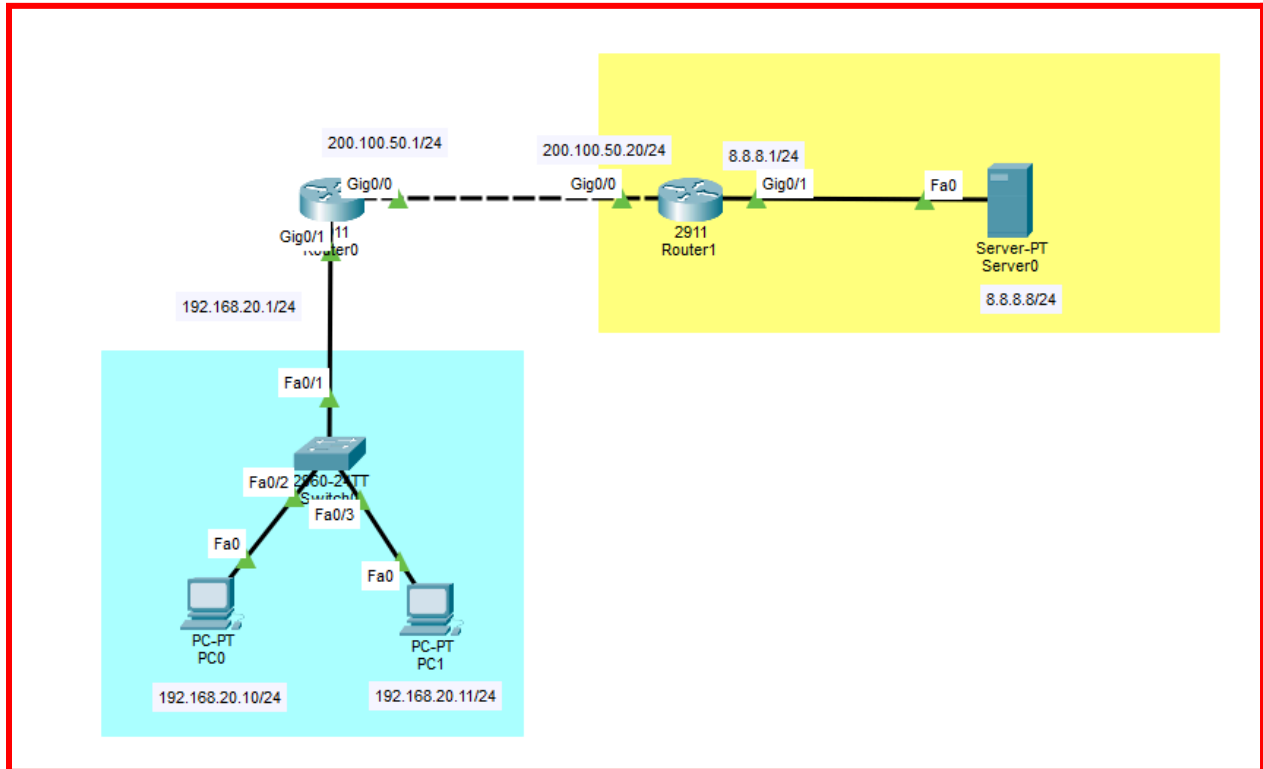
Step 3: Verifying Connections

To test the setup, go to a PC in the public network and attempt to access the private server by typing its public mapping (30.30.30.1) into a web browser. If configured correctly, the webpage will load and display the details you edited. Next, try to send a message from PC1 to PC2 directly, it should fail every time since devices on a public network should not be able to communicate with devices on the internal private network directly. Next, send a packet from PC2 to PC1, this should be sent successfully because it's an inside ip trying to communicate with the outside (this will be done through the router). Lastly, click on Inspect then click on Router 1 then click on the NAT table. You should be able to see a table showing the IPs in the internal and external network, indicating that NAT was set up correctly.



Task 2: Dynamic NAT (10 marks)

The following are the steps, topology and IP addresses you need to use for this task.



Step 1: Setting Up the Network Topology

- Place the following devices in the workspace:
 - 2 Routers (2911)
 - 1 Switch (2960)
 - 2 PCs (PC-PT)
 - 1 Server (Server-PT)

Connect all the devices to the appropriate ports as labelled in the diagram provided above.

Step 2: Assign IP Addresses

Set IPs via Desktop > IP Configuration:

- **PC0**
 - IP Address: 192.168.20.10



- Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.20.1
- **PC1**
 - IP Address: 192.168.20.11
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.20.1
- **Server**
 - IP Address: 8.8.8.8
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 8.8.8.1

Set IPs for the Routers by going to Config > Select an interface to configure

- **Router 0 (gig0/0)**
 - Turn the port status on
 - IP Address: 200.100.50.1
 - Subnet Mask: 255.255.255.0
- **Router 0 (gig0/1)**
 - Turn the port status on
 - IP Address: 192.168.20.1
 - Subnet Mask: 255.255.255.0
- **Router 1 (gig0/0)**
 - Turn the port status on
 - IP Address: 200.100.50.20
 - Subnet Mask: 255.255.255.0
- **Router 1 (gig0/1)**
 - Turn the port status on
 - IP Address: 8.8.8.1
 - Subnet Mask: 255.255.255.0

Step 3: Configuring OSPF Routing

To enable routing in your network and tell the router where to direct traffic, this task requires the configuration of a routing protocol (OSPF in this case). To carry this out, simply run the following commands on **Router 0**.

```
Router(config-if) #
Router(config-if) #exit
Router(config) #exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#config t
```



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```
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 45
Router(config-router)#router-id 4.3.2.1
Router(config-router)#network 200.100.50.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
00:13:39: %OSPF-5-ADJCHG: Process 45, Nbr 5.1.2.4 on GigabitEthernet0/0 from
LOADING to FULL, Loading Done
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Next, follow the same steps in **Router1**.

```
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 45
Router(config-router)#router-id 5.1.2.4
Router(config-router)#
Router(config-router)#network 8.8.8.0 0.0.0.255 area 0
Router(config-router)#network 200.100.50.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
00:13:36: %OSPF-5-ADJCHG: Process 45, Nbr 4.3.2.1 on GigabitEthernet0/0 from
LOADING to FULL, Loading Done
```

Step 4: Configuring Dynamic NAT

To enable dynamic NAT in your network, you will need to create an Access Control List (ACL). This list will tell the router which IPs from the internal network are allowed to use IPs from the Pool. To carry this out, simply run the following commands on Router 0.

```
Router#
Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
```



```
Router(config)#access-list 15 permit 192.168.20.0 0.0.0.255 //creating the ACL
Router(config)#ip nat pool DYNAT 200.100.50.1 200.100.50.10 netmask
255.255.255.0 //creating the NAT pool
Router(config)#ip nat inside source list 15 pool DYNAT //binding ACL to the pool
Router(config)#int gig0/0 //defining the NAT interfaces (what's inside and
what's outside the network)
Router(config-if)#ip nat outside
Router(config-if)#int gig0/1
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 5: Verifying Connectivity

To verify that dynamic NAT has been implemented, do the following.

On PC0:

- Open Command Prompt
- Run the command ping 8.8.8.8
- Note:
 - First ping may fail (ARP process)
 - Subsequent pings should succeed

On PC1:

- Open Command Prompt
- Run the command ping 8.8.8.8
- Note:
 - First ping may fail (ARP process)
 - Subsequent pings should succeed

On Router0:

- Open the CLI
- Run the command show ip nat translations
- You should see private IPs mapped to public IPs (2 different IPs, one that was given to PC0 and one that was given to PC1)



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Two different IPs

Submission Guidelines (5 Marks):

- Submit a single PDF document containing all tasks and the CISCO packet file(s) to the GCR, **also ensure your file name is visible in all pictures and contains your roll number.**
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1.**
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **L_i231234_Lab1. (for multiple files add the task number at the end)**
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.
- **In case a demo is conducted, there will be deductions based on your answers to the questions asked.**

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.



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Sample screenshot:

